MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

SECOND SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG2MDCSTA100 – Time Series Methods and their Applications

Duration: 1Hour Maximum Marks: 35

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

*Answer **All** Questions.*

*Each question carries **2** marks*

1. What is the primary characteristic of a time series data set? [Remember][CO1]
   (a) Data points are not related (b) Random data
   (c) Data is collected over time (d) None of these

2. Which of the following is not a time series plot? [Analyse][CO1]
   (a) Scatter plot (b) Month plot
   (c) Histogram (d) Season plot

3. Which of the following is a component of a time series? [Understand][CO1]
   (a) Trend (b) Seasonality
   (c) Irregularity (d) All of the above

4. The component of a time series that reflects long-term growth or decline is known as: [Remember][CO1]
   (a) Seasonal component (b) Trend component
   (c) Irregular component (d) Cyclical component

5. Identify the properties of a white noise model. [Skill][CO2]
   (a) Independent (b) Identical
   (c) Pattern free (d) All of the above

6. How would you make a random walk time series stationary? [Skill][CO2]
   (a) By taking the square of each observation.
   (b) By taking the first difference of the series.
   (c) By adding a constant to each observation.
   (d) By multiplying each observation by a constant.

7. Correlation coefficient is used to determine: [Interest][CO3]
   (a) A specific value of the y-variable given a specific value of the x-variable
   (b) A specific value of the x-variable given a specific value of the y-variable
   (c) The strength of the relationship between the x and y variables
   (d) None of these

8. What is the range of values that the auto correlation coefficient can take? [Interest][CO1, CO3]
   (a) -1 to 0 (b) 0 to 1
   (b) Any real number (d) -1 to 1

9. Which of the following is the possible cause of auto correlation in the time series data? [Interest] [CO1, CO3]

(a) Seasonality (b) Trend
(c) Measurement Error (d) All of the above

10. Identify the statistical tool for measuring the significance of correlation between a variable and the its lagged values. [Interest][CO1, CO3]

(a) PACF (b) Scatter plot
(b) Season plot (d) Month plot

(2 x 10 = 20)

Part B

Short Essay Type questions

*Answer any **3** questions.*

*Each question carries **5** marks.*

11. What are basic assumptions of time series? [Remember][CO1]
12. Briefly explain different methods of measurement of trend. [Skill][CO1, CO2]
13. Introduce the term stationary process. [Skill][CO2]
14. How would you construct a scatter plot? [Interest][CO3]
15. Comment on auto regressive models. [Interest][CO1, CO3]

(5 x 3 = 15)